CB.PS.I5DAS22006
AADARSH S NAIR

Untitled7

March 23, 2025

[176]:
```
import pandas as pd
import numpy as np
```

[177]:
```
df = pd.read_csv("laptop_data.csv")
df.head()
```

[177]:

	Unnamed: 0	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price
0	0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37kg	71378.6832
1	1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34kg	47895.5232
2	2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8GB	256GB SSD	Intel HD Graphics 620	No OS	1.86kg	30636.0000
3	3	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x1800	Intel Core i7 2.7GHz	16GB	512GB SSD	AMD Radeon Pro 455	macOS	1.83kg	135195.3360
4	4	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 3.1GHz	8GB	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37kg	96095.8080

[178]:
```
df.shape
```

[178]: (1303, 12)

[179]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Unnamed: 0        1303 non-null   int64
```

```
 1   Company           1303 non-null   object
 2   TypeName          1303 non-null   object
 3   Inches            1303 non-null   float64
 4   ScreenResolution  1303 non-null   object
 5   Cpu               1303 non-null   object
 6   Ram               1303 non-null   object
 7   Memory            1303 non-null   object
 8   Gpu               1303 non-null   object
 9   OpSys             1303 non-null   object
 10  Weight            1303 non-null   object
 11  Price             1303 non-null   float64
dtypes: float64(2), int64(1), object(9)
memory usage: 122.3+ KB
```

[180]:
```
df.isnull().sum()
```

[180]:
```
Unnamed: 0          0
Company             0
TypeName            0
Inches              0
ScreenResolution    0
Cpu                 0
Ram                 0
Memory              0
Gpu                 0
OpSys               0
Weight              0
Price               0
dtype: int64
```

[181]:
```
df.duplicated().sum()
```

[181]:
```
0
```

[182]:
```
df.drop(columns=["Unnamed: 0"],inplace=True)
```

[183]:
```
df.head()
```

[183]:

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37kg	71378.6832
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34kg	47895.5232
2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8GB	256GB SSD	Intel HD Graphics 620	No OS	1.86kg	30636.0000
3	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x1800	Intel Core i7 2.7GHz	16GB	512GB SSD	AMD Radeon Pro 455	macOS	1.83kg	135195.3360
4	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 3.1GHz	8GB	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37kg	96095.8080

```
2  Intel Core i5 7200U 2.5GHz   8GB            256GB SSD
3         Intel Core i7 2.7GHz  16GB            512GB SSD
4         Intel Core i5 3.1GHz   8GB            256GB SSD

                             Gpu  OpSys  Weight        Price
0  Intel Iris Plus Graphics 640  macOS  1.37kg   71378.6832
1        Intel HD Graphics 6000  macOS  1.34kg   47895.5232
2         Intel HD Graphics 620  No OS  1.86kg   30636.0000
3            AMD Radeon Pro 455  macOS  1.83kg  135195.3360
4  Intel Iris Plus Graphics 650  macOS  1.37kg   96095.8080
```

[184]:
```
df['Ram']=df['Ram'].str.replace('GB','')
df['Weight']=df['Weight'].str.replace('kg','')
```

[185]:
```
df.head()
```

[185]:
```
   Company   TypeName  Inches                    ScreenResolution  \
0   Apple  Ultrabook    13.3  IPS Panel Retina Display 2560x1600
1   Apple  Ultrabook    13.3                            1440x900
2      HP   Notebook    15.6                   Full HD 1920x1080
3   Apple  Ultrabook    15.4  IPS Panel Retina Display 2880x1800
4   Apple  Ultrabook    13.3  IPS Panel Retina Display 2560x1600

                          Cpu Ram               Memory  \
0        Intel Core i5 2.3GHz   8            128GB SSD
1        Intel Core i5 1.8GHz   8  128GB Flash Storage
2  Intel Core i5 7200U 2.5GHz   8            256GB SSD
3        Intel Core i7 2.7GHz  16            512GB SSD
4        Intel Core i5 3.1GHz   8            256GB SSD

                             Gpu  OpSys Weight        Price
0  Intel Iris Plus Graphics 640  macOS   1.37   71378.6832
1        Intel HD Graphics 6000  macOS   1.34   47895.5232
2         Intel HD Graphics 620  No OS   1.86   30636.0000
3            AMD Radeon Pro 455  macOS   1.83  135195.3360
4  Intel Iris Plus Graphics 650  macOS   1.37   96095.8080
```

[186]:
```
df['Ram']=df['Ram'].astype('int32')
df['Weight']=df['Weight'].astype('float32')
```

[187]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 11 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Company           1303 non-null   object
```

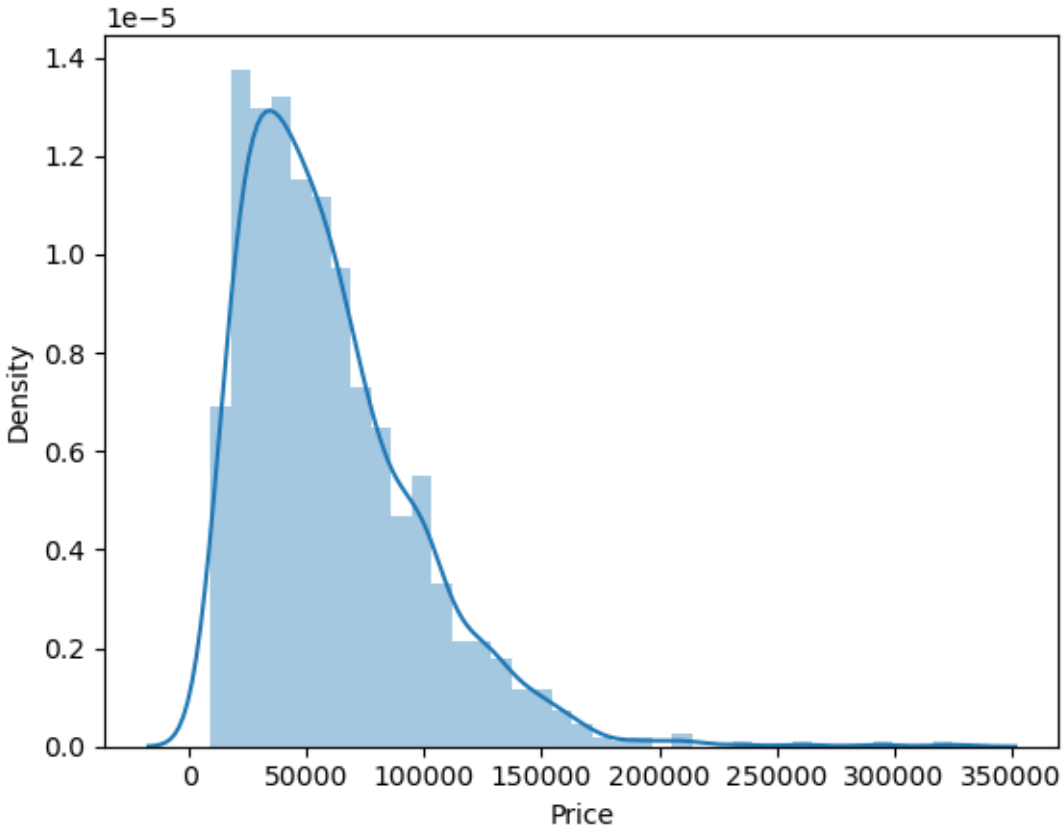
```
 1   TypeName          1303 non-null   object
 2   Inches            1303 non-null   float64
 3   ScreenResolution  1303 non-null   object
 4   Cpu               1303 non-null   object
 5   Ram               1303 non-null   int32
 6   Memory            1303 non-null   object
 7   Gpu               1303 non-null   object
 8   OpSys             1303 non-null   object
 9   Weight            1303 non-null   float32
 10  Price             1303 non-null   float64
dtypes: float32(1), float64(2), int32(1), object(7)
memory usage: 101.9+ KB
```

[188]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

[189]:
```
sns.distplot(df["Price"])
```

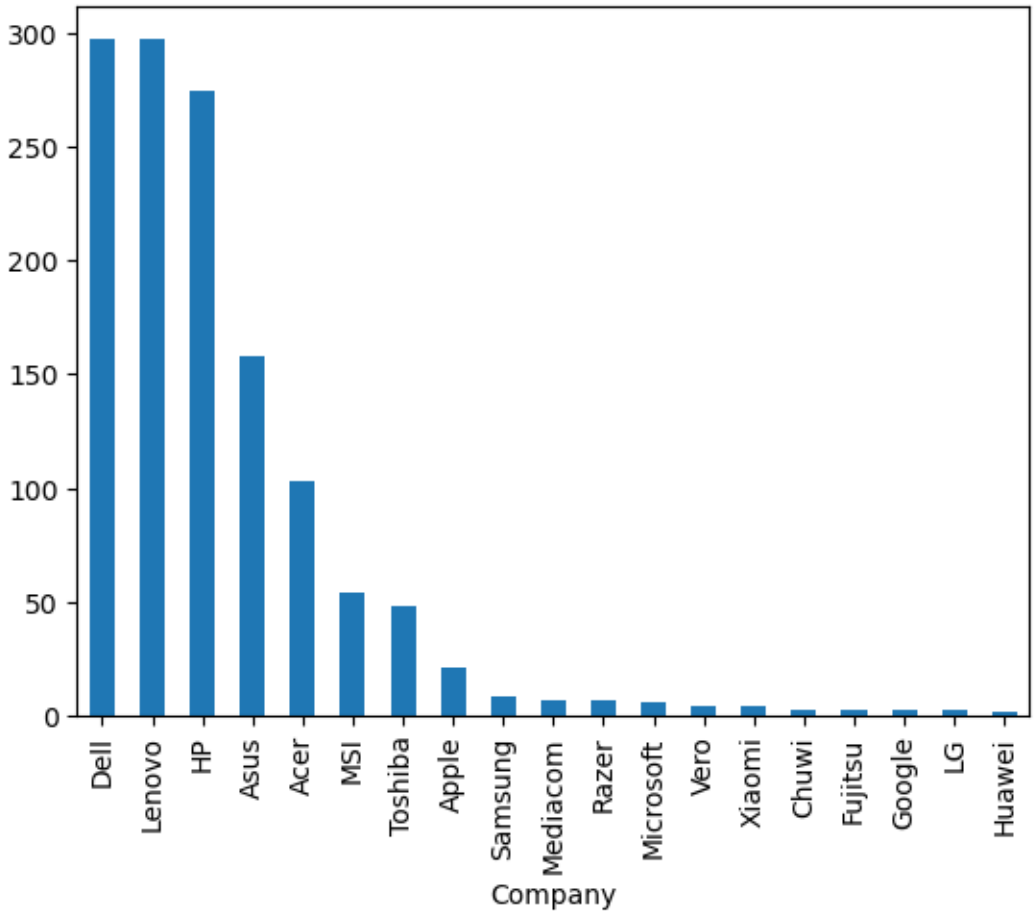
```
/tmp/ipykernel_75163/941010651.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df["Price"])
```

[189]: <AxesSubplot: xlabel='Price', ylabel='Density'>

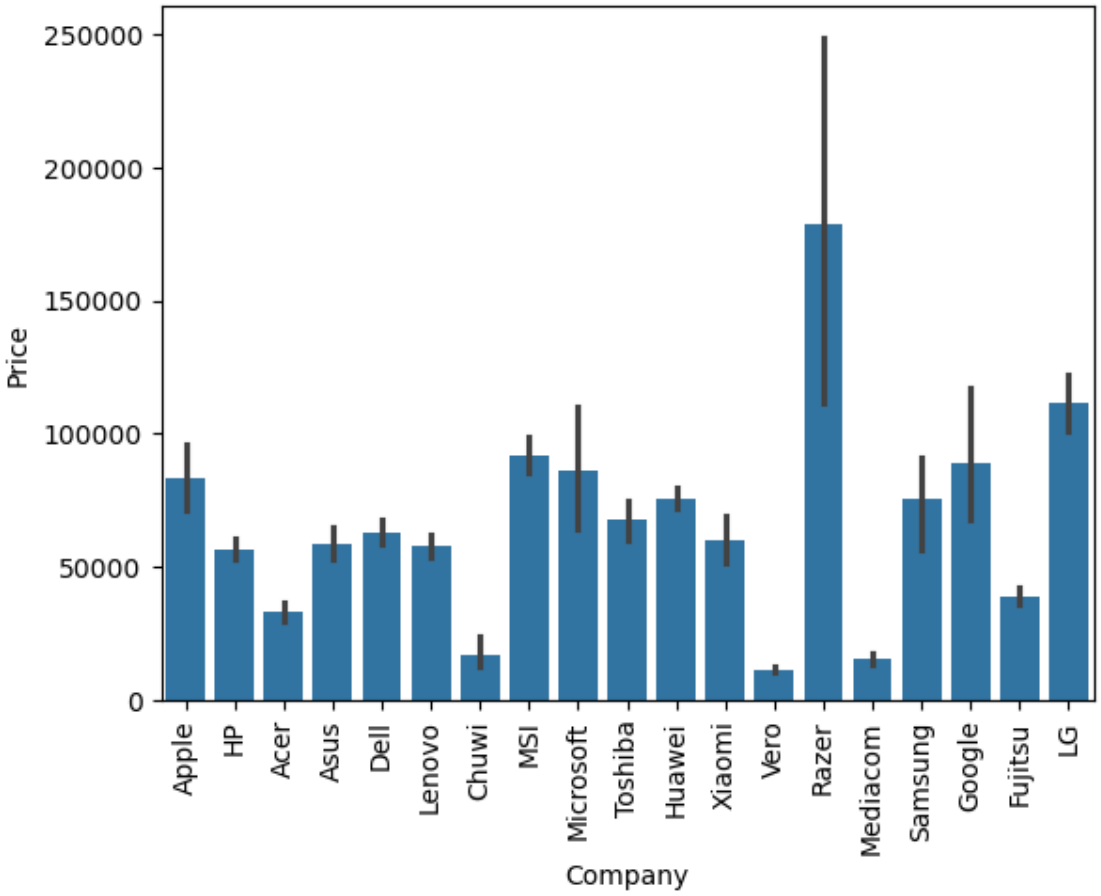


[190]:
```
df['Company'].value_counts().plot(kind='bar')
```

[190]: <AxesSubplot: xlabel='Company'>

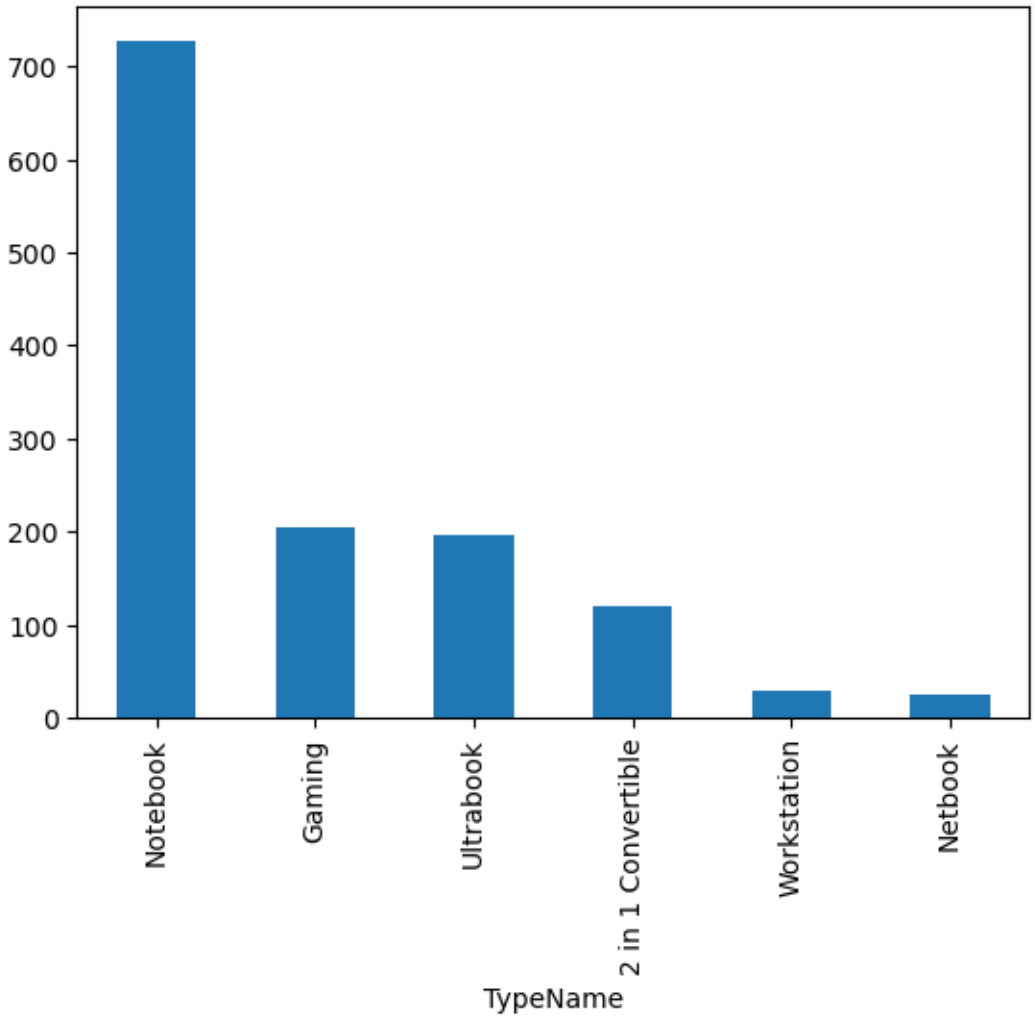


[191]:
```
sns.barplot(x=df['Company'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```

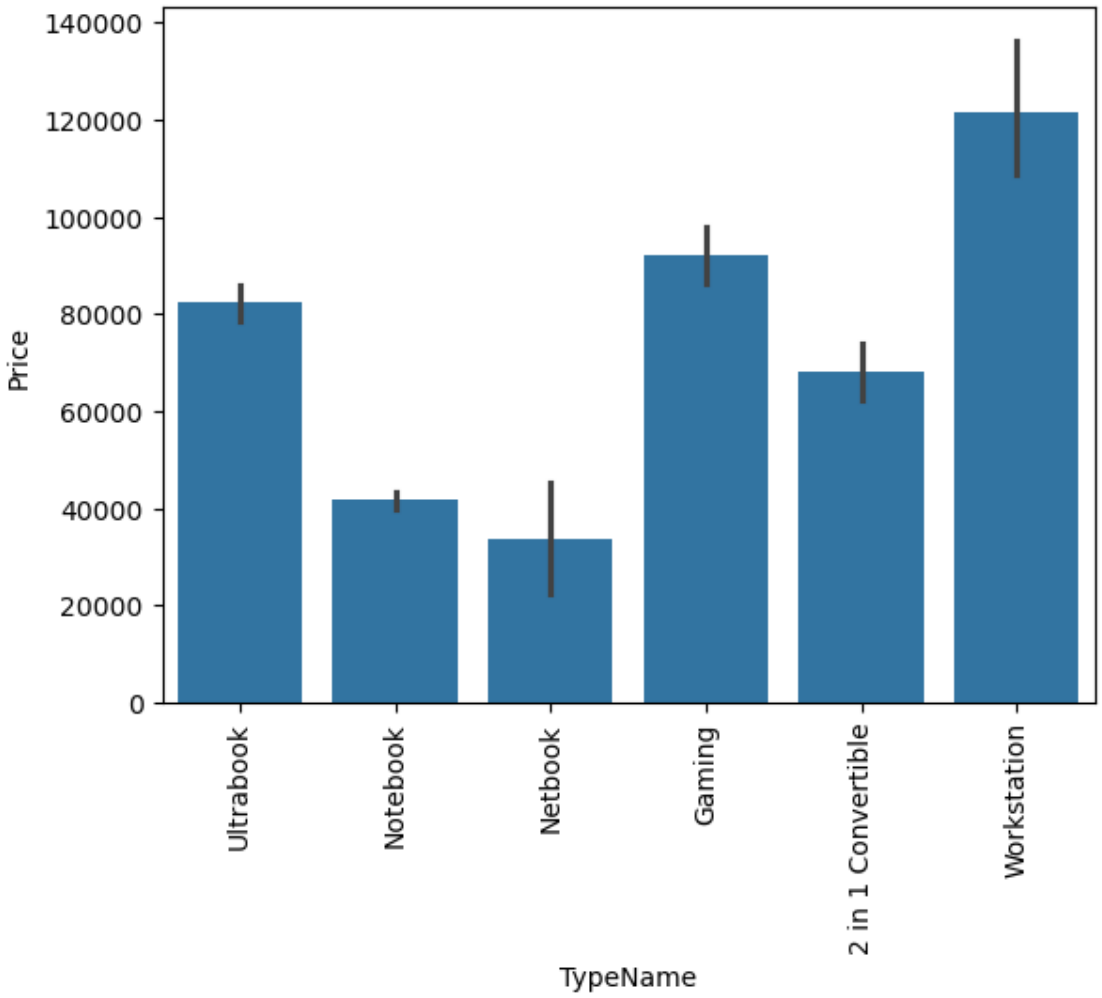


[192]:
```
df['TypeName'].value_counts().plot(kind='bar')
```

[192]: <AxesSubplot: xlabel='TypeName'>



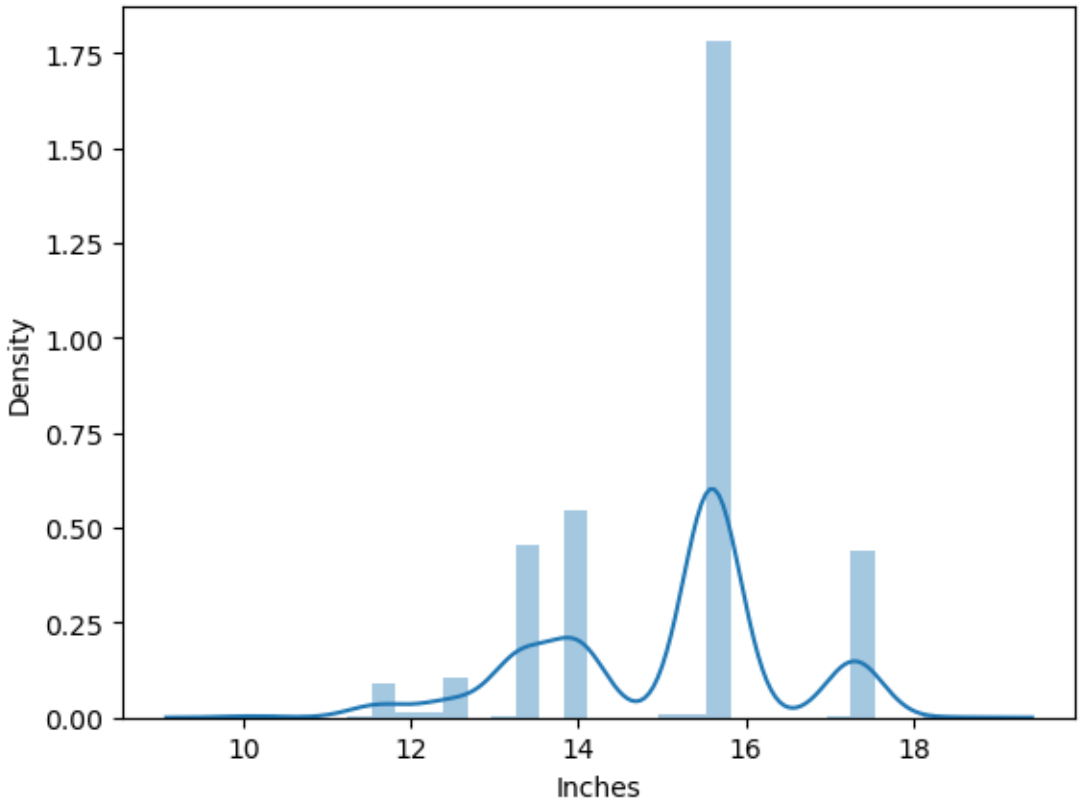
[193]:
```
sns.barplot(x=df['TypeName'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```



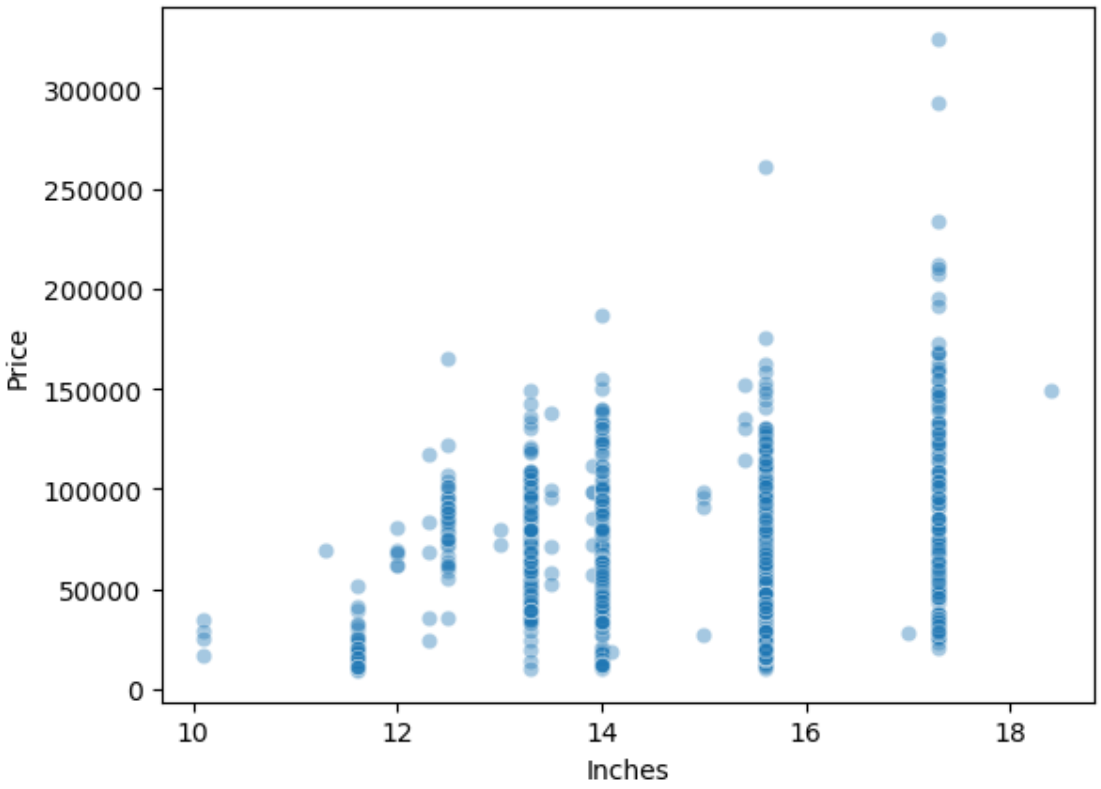
[194]:
```
sns.distplot(df["Inches"])
```

```
/tmp/ipykernel_75163/480038204.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df["Inches"])
```

[194]: <AxesSubplot: xlabel='Inches', ylabel='Density'>



[195]:
```
sns.scatterplot(x=df['Inches'],y=df['Price'],alpha=0.4)
plt.show()
```



[196]:
```
df['ScreenResolution'].value_counts()
```

[196]:
```
ScreenResolution
Full HD 1920x1080                                507
1366x768                                         281
IPS Panel Full HD 1920x1080                      230
IPS Panel Full HD / Touchscreen 1920x1080         53
Full HD / Touchscreen 1920x1080                   47
1600x900                                          23
Touchscreen 1366x768                              16
Quad HD+ / Touchscreen 3200x1800                  15
IPS Panel 4K Ultra HD 3840x2160                   12
IPS Panel 4K Ultra HD / Touchscreen 3840x2160     11
4K Ultra HD / Touchscreen 3840x2160               10
IPS Panel 1366x768                                 7
Touchscreen 2560x1440                              7
4K Ultra HD 3840x2160                              7
IPS Panel Retina Display 2304x1440                 6
IPS Panel Retina Display 2560x1600                 6
Touchscreen 2256x1504                              6
IPS Panel Quad HD+ / Touchscreen 3200x1800         6
```

```
IPS Panel Touchscreen 2560x1440                    5
IPS Panel Retina Display 2880x1800                 4
1440x900                                           4
IPS Panel Touchscreen 1920x1200                    4
IPS Panel 2560x1440                                4
IPS Panel Quad HD+ 2560x1440                       3
IPS Panel Touchscreen 1366x768                     3
Quad HD+ 3200x1800                                 3
1920x1080                                          3
2560x1440                                          3
Touchscreen 2400x1600                              3
IPS Panel Quad HD+ 3200x1800                       2
IPS Panel Full HD 2160x1440                        2
IPS Panel Touchscreen / 4K Ultra HD 3840x2160      2
IPS Panel Full HD 1366x768                         1
Touchscreen / Quad HD+ 3200x1800                   1
IPS Panel Retina Display 2736x1824                 1
IPS Panel Full HD 2560x1440                        1
IPS Panel Full HD 1920x1200                        1
Touchscreen / Full HD 1920x1080                    1
Touchscreen / 4K Ultra HD 3840x2160                1
IPS Panel Touchscreen 2400x1600                    1
Name: count, dtype: int64
```

[197]:
```
df['Touchscreen']=df['ScreenResolution'].apply(lambda x:1 if 'Touchscreen' in x␣
↪else 0)
```

[198]:
```
df.sample(5)
```

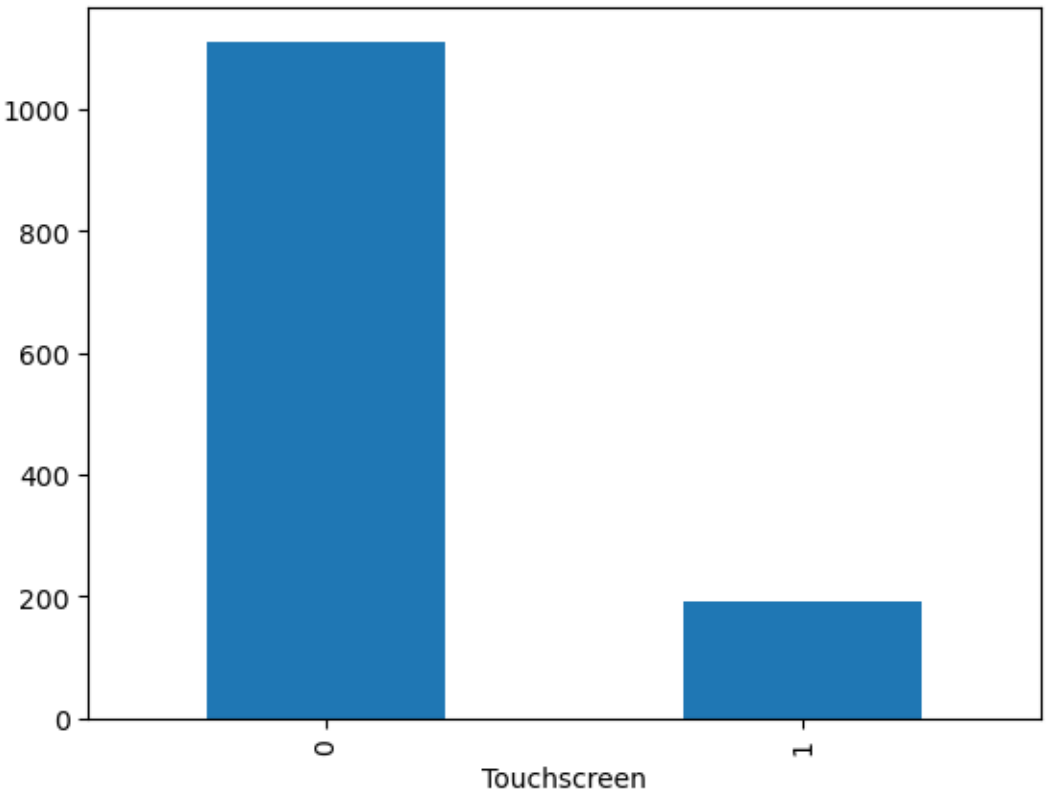
[198]:

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen
428	HP	Gaming	17.3	Full HD 1920x1080	Intel Core i7 7700HQ 2.8GHz	12	256GB SSD + 1TB HDD	Nvidia GeForce GTX 1070	Windows 10	3.35	106506.72	0
231	HP	Notebook	15.6	1366x768	AMD E-Series 9000e 1.5GHz	4	500GB HDD	AMD Radeon R2	Windows 10	2.10	17582.40	0
813	Dell	Notebook	15.6	Full HD 1920x1080	Intel Core i7 7500U 2.7GHz	8	1TB HDD	Nvidia GeForce GT 940MX	Windows 10	1.98	51202.08	0
566	Dell	Notebook	15.6	1366x768	Intel Core i5 7300U 2.6GHz	4	500GB HDD					
1173	Lenovo	Notebook	15.6	1366x768	Intel Core i5 6200U 2.3GHz	4	500GB HDD					

```
566     Intel HD Graphics 620  Windows 10    1.93   51095.52           0
1173    Intel HD Graphics 520       No OS    2.10   21205.44           0
```

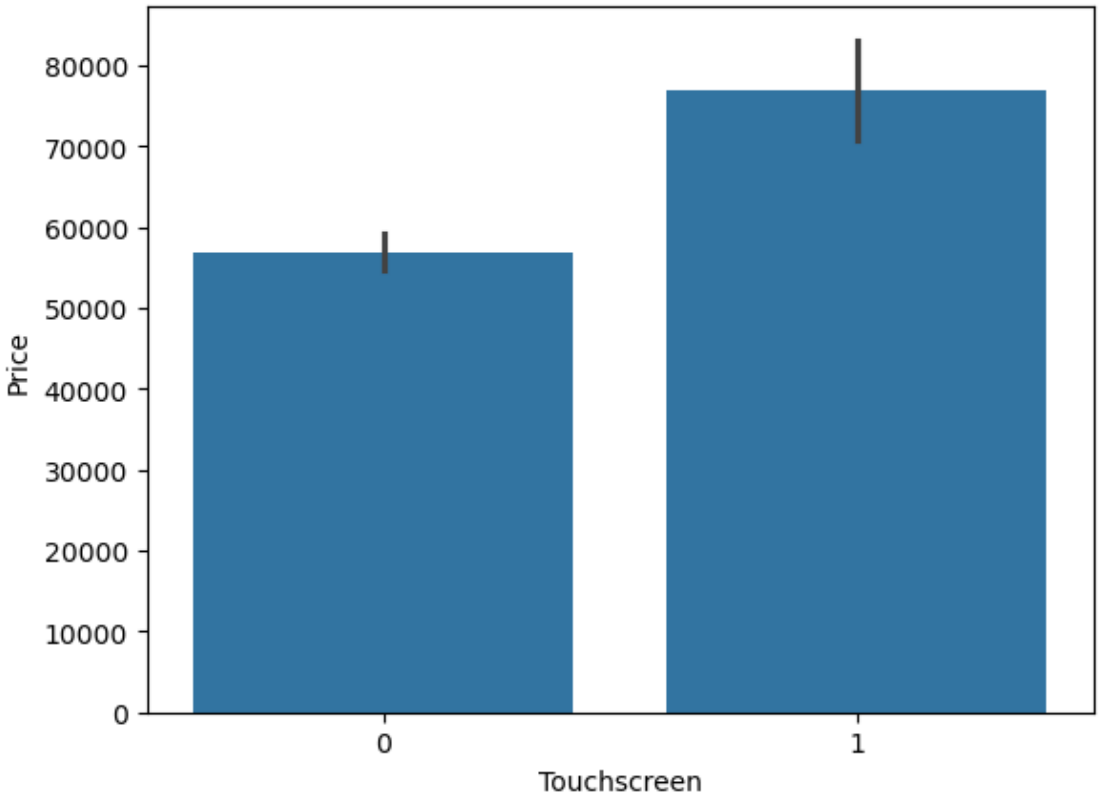
[199]:
```
df['Touchscreen'].value_counts().plot(kind='bar')
```

[199]:
```
<AxesSubplot: xlabel='Touchscreen'>
```



[200]:
```
sns.barplot(x=df['Touchscreen'],y=df['Price'])
```

[200]:
```
<AxesSubplot: xlabel='Touchscreen', ylabel='Price'>
```



[201]:
```
df['Ips']=df['ScreenResolution'].apply(lambda x:1 if 'IPS' in x else 0)
```

[202]:
```
df.head()
```

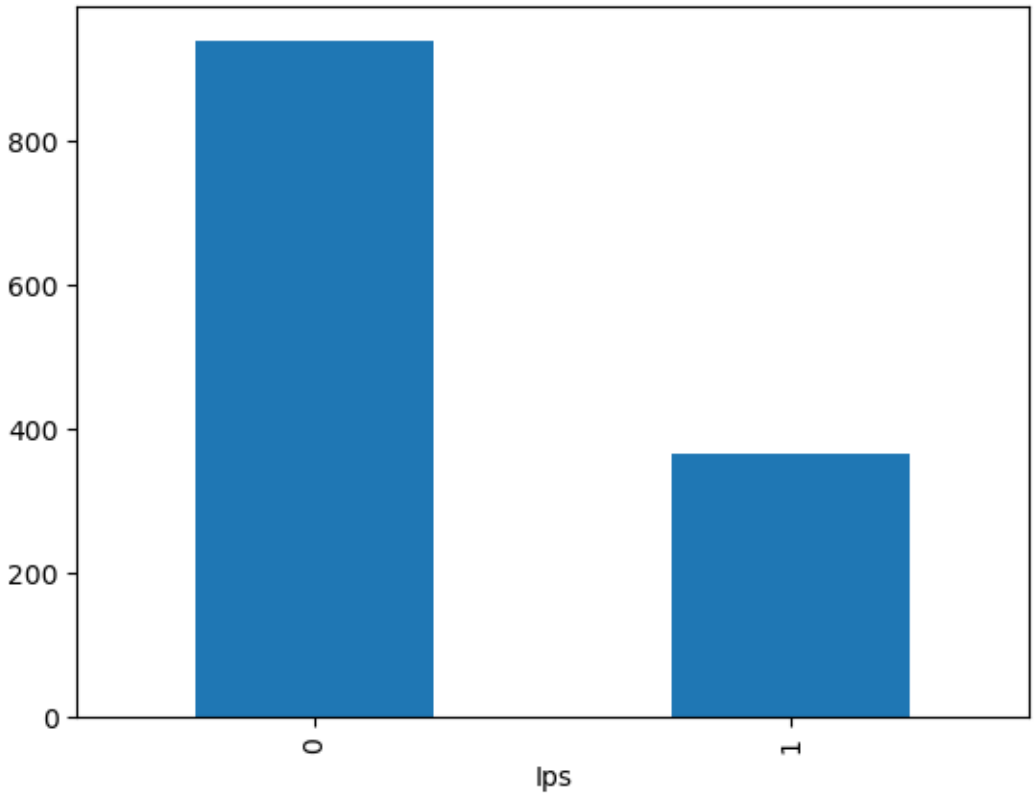
[202]:

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0
2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8	256GB SSD						
3	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x1800	Intel Core i7 2.7GHz	16	512GB SSD						
4	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 3.1GHz	8	256GB SSD						

```
2          Intel HD Graphics 620  No OS    1.86   30636.0000           0    0
3             AMD Radeon Pro 455  macOS    1.83  135195.3360           0    1
4  Intel Iris Plus Graphics 650  macOS    1.37   96095.8080           0    1
```

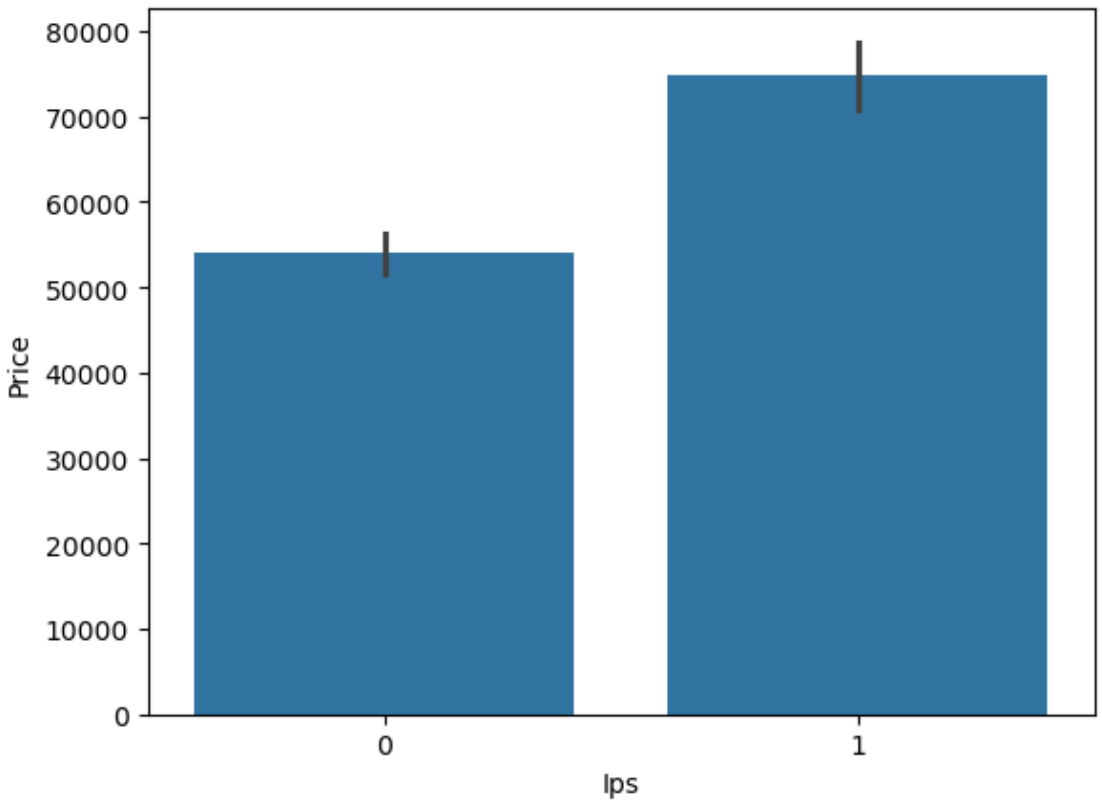
[203]:
```
df['Ips'].value_counts().plot(kind='bar')
```

[203]: <AxesSubplot: xlabel='Ips'>



[204]:
```
sns.barplot(x=df['Ips'],y=df['Price'])
```

[204]: <AxesSubplot: xlabel='Ips', ylabel='Price'>



[205]:
```
new = df['ScreenResolution'].str.split('x',n=1,expand=True)
```

[206]:
```
df['X_res'] = new[0]
df['Y_res'] = new[1]
```

[207]:
```
df.head()
```

[207]:

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8	128GB SSD
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8	128GB Flash Storage
2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8	256GB SSD
3	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x1800	Intel Core i7 2.7GHz	16	512GB SSD
4	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 3.1GHz	8	256GB SSD

```
                            Gpu  OpSys  Weight        Price  Touchscreen  Ips  \
0  Intel Iris Plus Graphics 640  macOS    1.37   71378.6832            0    1
1        Intel HD Graphics 6000  macOS    1.34   47895.5232            0    0
2         Intel HD Graphics 620  No OS    1.86   30636.0000            0    0
3            AMD Radeon Pro 455  macOS    1.83  135195.3360            0    1
4  Intel Iris Plus Graphics 650  macOS    1.37   96095.8080            0    1

                           X_res Y_res
0  IPS Panel Retina Display 2560  1600
1                        1440   900
2                  Full HD 1920  1080
3  IPS Panel Retina Display 2880  1800
4  IPS Panel Retina Display 2560  1600
```

[208]:
```
df['X_res'].str.replace(',','').str.findall(r'(\d+.?\d+)').apply(lambda x:x[0])
```

[208]:
```
0       2560
1       1440
2       1920
3       2880
4       2560
       ...
1298    1920
1299    3200
1300    1366
1301    1366
1302    1366
Name: X_res, Length: 1303, dtype: object
```

[209]:
```
df['X_res']=df['X_res'].str.replace(',','').str.findall(r'(\d+.?\d+)').
↪apply(lambda x:x[0])
```

[210]:
```
df.head()
```

[210]:
```
  Company   TypeName  Inches                    ScreenResolution  \
0   Apple  Ultrabook    13.3  IPS Panel Retina Display 2560x1600
1   Apple  Ultrabook    13.3                            1440x900
2      HP   Notebook    15.6                  Full HD 1920x1080
3   Apple  Ultrabook    15.4  IPS Panel Retina Display 2880x1800
4   Apple  Ultrabook    13.3  IPS Panel Retina Display 2560x1600

                          Cpu  Ram               Memory  \
0        Intel Core i5 2.3GHz    8            128GB SSD
1        Intel Core i5 1.8GHz    8  128GB Flash Storage
2  Intel Core i5 7200U 2.5GHz    8            256GB SSD
3        Intel Core i7 2.7GHz   16            512GB SSD
4        Intel Core i5 3.1GHz    8            256GB SSD
```

```
                            Gpu  OpSys  Weight        Price  Touchscreen  Ips  \
0  Intel Iris Plus Graphics 640  macOS    1.37   71378.6832            0    1
1        Intel HD Graphics 6000  macOS    1.34   47895.5232            0    0
2         Intel HD Graphics 620  No OS    1.86   30636.0000            0    0
3            AMD Radeon Pro 455  macOS    1.83  135195.3360            0    1
4  Intel Iris Plus Graphics 650  macOS    1.37   96095.8080            0    1

  X_res Y_res
0  2560  1600
1  1440   900
2  1920  1080
3  2880  1800
4  2560  1600
```

[ ]:

[212]:
```
df.head()
```

[212]:
```
  Company   TypeName  Inches                    ScreenResolution  \
0   Apple  Ultrabook    13.3  IPS Panel Retina Display 2560x1600
1   Apple  Ultrabook    13.3                            1440x900
2      HP   Notebook    15.6                  Full HD 1920x1080
3   Apple  Ultrabook    15.4  IPS Panel Retina Display 2880x1800
4   Apple  Ultrabook    13.3  IPS Panel Retina Display 2560x1600

                          Cpu  Ram               Memory  \
0        Intel Core i5 2.3GHz    8            128GB SSD
1        Intel Core i5 1.8GHz    8  128GB Flash Storage
2  Intel Core i5 7200U 2.5GHz    8            256GB SSD
3        Intel Core i7 2.7GHz   16            512GB SSD
4        Intel Core i5 3.1GHz    8            256GB SSD

                            Gpu  OpSys  Weight        Price  Touchscreen  Ips  \
0  Intel Iris Plus Graphics 640  macOS    1.37   71378.6832            0    1
1        Intel HD Graphics 6000  macOS    1.34   47895.5232            0    0
2         Intel HD Graphics 620  No OS    1.86   30636.0000            0    0
3            AMD Radeon Pro 455  macOS    1.83  135195.3360            0    1
4  Intel Iris Plus Graphics 650  macOS    1.37   96095.8080            0    1

  X_res Y_res
0  2560  1600
1  1440   900
2  1920  1080
3  2880  1800
4  2560  1600
```

[213]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Company           1303 non-null   object
 1   TypeName          1303 non-null   object
 2   Inches            1303 non-null   float64
 3   ScreenResolution  1303 non-null   object
 4   Cpu               1303 non-null   object
 5   Ram               1303 non-null   int32
 6   Memory            1303 non-null   object
 7   Gpu               1303 non-null   object
 8   OpSys             1303 non-null   object
 9   Weight            1303 non-null   float32
 10  Price             1303 non-null   float64
 11  Touchscreen       1303 non-null   int64
 12  Ips               1303 non-null   int64
 13  X_res             1303 non-null   object
 14  Y_res             1303 non-null   object
dtypes: float32(1), float64(2), int32(1), int64(2), object(9)
memory usage: 142.6+ KB
```

[214]:
```
df['X_res']=df['X_res'].astype('int32')
df['Y_res']=df['Y_res'].astype('int32')
```

[215]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1303 entries, 0 to 1302
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Company           1303 non-null   object
 1   TypeName          1303 non-null   object
 2   Inches            1303 non-null   float64
 3   ScreenResolution  1303 non-null   object
 4   Cpu               1303 non-null   object
 5   Ram               1303 non-null   int32
 6   Memory            1303 non-null   object
 7   Gpu               1303 non-null   object
 8   OpSys             1303 non-null   object
 9   Weight            1303 non-null   float32
 10  Price             1303 non-null   float64
 11  Touchscreen       1303 non-null   int64
 12  Ips               1303 non-null   int64
```

```
 13  X_res             1303 non-null   int32
 14  Y_res             1303 non-null   int32
dtypes: float32(1), float64(2), int32(3), int64(2), object(7)
memory usage: 132.5+ KB
```

[216]:
```
df.select_dtypes(include='number').corr()['Price']
```

[216]:
```
Inches         0.068197
Ram            0.743007
Weight         0.210370
Price          1.000000
Touchscreen    0.191226
Ips            0.252208
X_res          0.556529
Y_res          0.552809
Name: Price, dtype: float64
```

[217]:
```
df['ppi']=np.sqrt((df['X_res']**2)+(df['Y_res']**2))/df['Inches'].
↪astype('float')
```

[218]:
```
df.head()
```

[218]:

	Company	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	X_res	Y_res	ppi
0	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	2560	1600	226.983005
1	Apple	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	1440	900	127.677940
2	HP	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8	256GB SSD	Intel HD Graphics 620	No OS	1.86	30636.0000	0	0	1920	1080	141.211998
3	Apple	Ultrabook	15.4	IPS Panel Retina Display 2880x1800	Intel Core i7 2.7GHz	16	512GB SSD	AMD Radeon Pro 455	macOS	1.83	135195.3360	0	1	2880	1800	220.534624
4	Apple	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 3.1GHz	8	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37	96095.8080	0	1	2560	1600	226.983005

```
4   2560   1600  226.983005
```

[219]:
```
df.select_dtypes(include='number').corr()['Price']
```

[219]:
```
Inches         0.068197
Ram            0.743007
Weight         0.210370
Price          1.000000
Touchscreen    0.191226
Ips            0.252208
X_res          0.556529
Y_res          0.552809
ppi            0.473487
Name: Price, dtype: float64
```

[220]:
```
df.drop(columns=["ScreenResolution"],inplace=True)
```

[221]:
```
df.head()
```

[221]:

	Company	TypeName	Inches	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	X_res	Y_res	ppi
0	Apple	Ultrabook	13.3	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	2560	1600	226.983005
1	Apple	Ultrabook	13.3	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	1440	900	127.677940
2	HP	Notebook	15.6	Intel Core i5 7200U 2.5GHz	8	256GB SSD	Intel HD Graphics 620	No OS	1.86	30636.0000	0	0	1920	1080	141.211998
3	Apple	Ultrabook	15.4	Intel Core i7 2.7GHz	16	512GB SSD	AMD Radeon Pro 455	macOS	1.83	135195.3360	0	1	2880	1800	220.534624
4	Apple	Ultrabook	13.3	Intel Core i5 3.1GHz	8	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37	96095.8080	0	1	2560	1600	226.983005

[222]:
```
df.drop(columns=["Inches","X_res","Y_res"],inplace=True)
```

[223]:
```
df.head()
```

[223]:

	Company	TypeName	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi
0	Apple	Ultrabook	Intel Core i5 2.3GHz	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	226.983005
1	Apple	Ultrabook	Intel Core i5 1.8GHz	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	127.677940
2	HP	Notebook	Intel Core i5 7200U 2.5GHz	8	256GB SSD	Intel HD Graphics 620	No OS	1.86	30636.0000	0	0	141.211998

```
3   Apple  Ultrabook        Intel Core i7 2.7GHz   16            512GB SSD
4   Apple  Ultrabook        Intel Core i5 3.1GHz    8            256GB SSD

                            Gpu  OpSys  Weight        Price  Touchscreen  Ips  \
0  Intel Iris Plus Graphics 640  macOS    1.37   71378.6832            0    1
1        Intel HD Graphics 6000  macOS    1.34   47895.5232            0    0
2         Intel HD Graphics 620  No OS    1.86   30636.0000            0    0
3            AMD Radeon Pro 455  macOS    1.83  135195.3360            0    1
4  Intel Iris Plus Graphics 650  macOS    1.37   96095.8080            0    1

          ppi
0  226.983005
1  127.677940
2  141.211998
3  220.534624
4  226.983005
```

[224]:
```
df['Cpu'].value_counts()
```

[224]:
```
Cpu
Intel Core i5 7200U 2.5GHz     190
Intel Core i7 7700HQ 2.8GHz    146
Intel Core i7 7500U 2.7GHz     134
Intel Core i7 8550U 1.8GHz      73
Intel Core i5 8250U 1.6GHz      72
                              ...
Intel Core M M7-6Y75 1.2GHz      1
Intel Core i5 7200U 2.50GHz      1
Intel Core i5 7200U 2.70GHz      1
Intel Core i5 7200U 2.7GHz       1
Intel Core i7 2.7GHz             1
Name: count, Length: 118, dtype: int64
```

[225]:
```
df['Cpu Name']=df['Cpu'].apply(lambda x:" ".join(x.split()[0:3]))
```

[226]:
```
df.head()
```

[226]:
```
  Company   TypeName                          Cpu  Ram               Memory  \
0   Apple  Ultrabook        Intel Core i5 2.3GHz    8            128GB SSD
1   Apple  Ultrabook        Intel Core i5 1.8GHz    8  128GB Flash Storage
2      HP   Notebook  Intel Core i5 7200U 2.5GHz    8            256GB SSD
3   Apple  Ultrabook        Intel Core i7 2.7GHz   16            512GB SSD
4   Apple  Ultrabook        Intel Core i5 3.1GHz    8            256GB SSD

                            Gpu  OpSys  Weight        Price  Touchscreen  Ips  \
0  Intel Iris Plus Graphics 640  macOS    1.37   71378.6832            0    1
1        Intel HD Graphics 6000  macOS    1.34   47895.5232            0    0
```

```
2         Intel HD Graphics 620  No OS    1.86   30636.0000            0    0
3            AMD Radeon Pro 455  macOS    1.83  135195.3360            0    1
4  Intel Iris Plus Graphics 650  macOS    1.37   96095.8080            0    1

          ppi       Cpu Name
0  226.983005  Intel Core i5
1  127.677940  Intel Core i5
2  141.211998  Intel Core i5
3  220.534624  Intel Core i7
4  226.983005  Intel Core i5
```

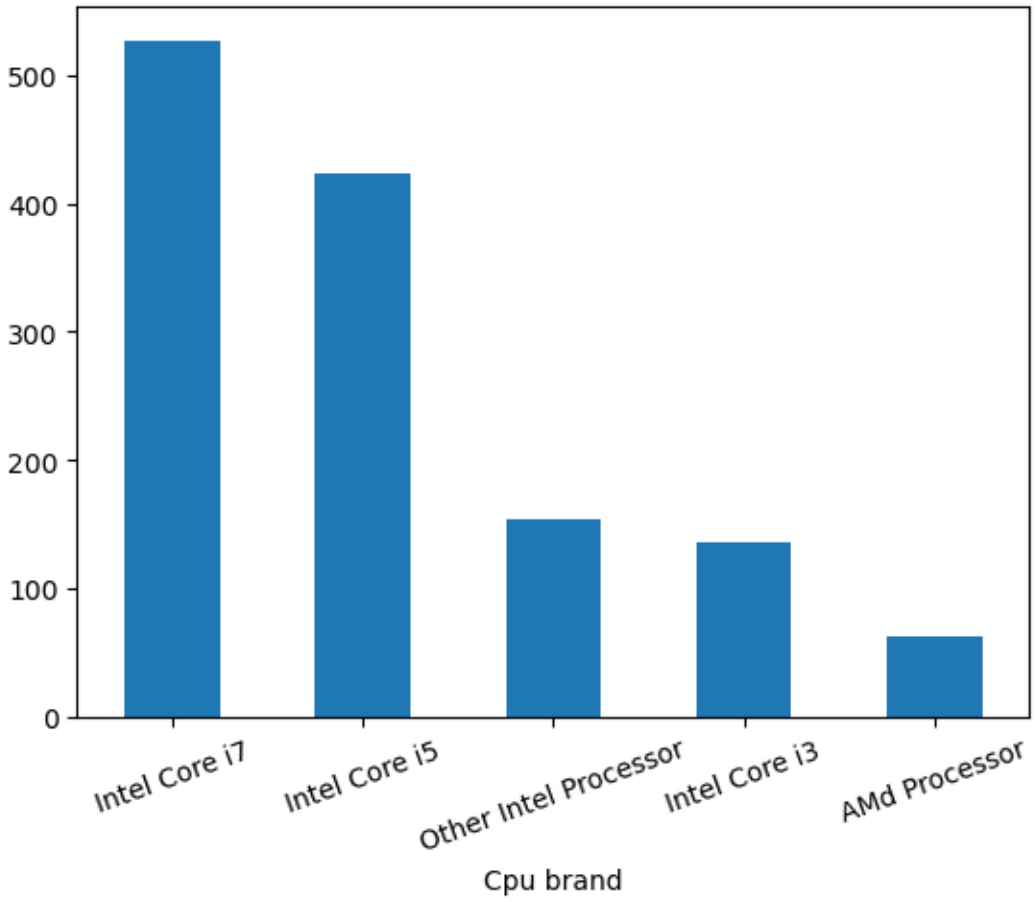
[227]:
```
def fetch_processor(text):
       if text == 'Intel Core i7' or text == 'Intel Core i5' or text == "Intel␣
↪Core i3":
           return text
       else:
           if text.split()[0] == 'Intel':
               return 'Other Intel Processor'
           else:
               return 'AMd Processor'
```

[228]:
```
df['Cpu brand'] = df['Cpu Name'].apply(fetch_processor)
```

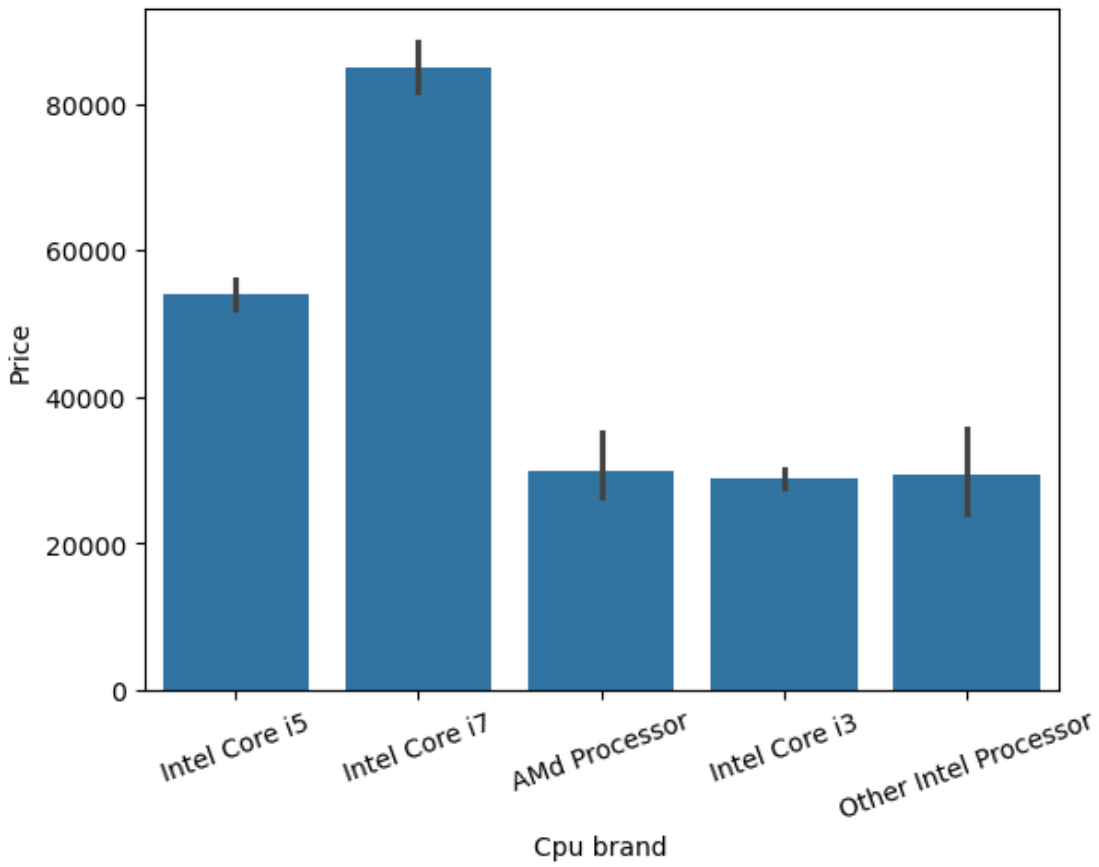
[229]:
```
df.head()
```

[229]:
```
  Company   TypeName                          Cpu  Ram               Memory  \
0   Apple  Ultrabook        Intel Core i5 2.3GHz    8            128GB SSD
1   Apple  Ultrabook        Intel Core i5 1.8GHz    8  128GB Flash Storage
2      HP   Notebook  Intel Core i5 7200U 2.5GHz    8            256GB SSD
3   Apple  Ultrabook        Intel Core i7 2.7GHz   16            512GB SSD
4   Apple  Ultrabook        Intel Core i5 3.1GHz    8            256GB SSD

                           Gpu  OpSys  Weight        Price  Touchscreen  Ips  \
0  Intel Iris Plus Graphics 640  macOS    1.37   71378.6832            0    1
1        Intel HD Graphics 6000  macOS    1.34   47895.5232            0    0
2         Intel HD Graphics 620  No OS    1.86   30636.0000            0    0
3            AMD Radeon Pro 455  macOS    1.83  135195.3360            0    1
4  Intel Iris Plus Graphics 650  macOS    1.37   96095.8080            0    1

          ppi       Cpu Name      Cpu brand
0  226.983005  Intel Core i5  Intel Core i5
1  127.677940  Intel Core i5  Intel Core i5
2  141.211998  Intel Core i5  Intel Core i5
3  220.534624  Intel Core i7  Intel Core i7
4  226.983005  Intel Core i5  Intel Core i5
```

[230]:
```
df['Cpu brand'].value_counts().plot(kind='bar')
plt.xticks(rotation=20)
plt.show()
```



[231]:
```
sns.barplot(x=df['Cpu brand'],y=df['Price'])
plt.xticks(rotation=20)
plt.show()
```



[232]:
```
df.drop(columns=['Cpu','Cpu Name'],inplace=True)
```

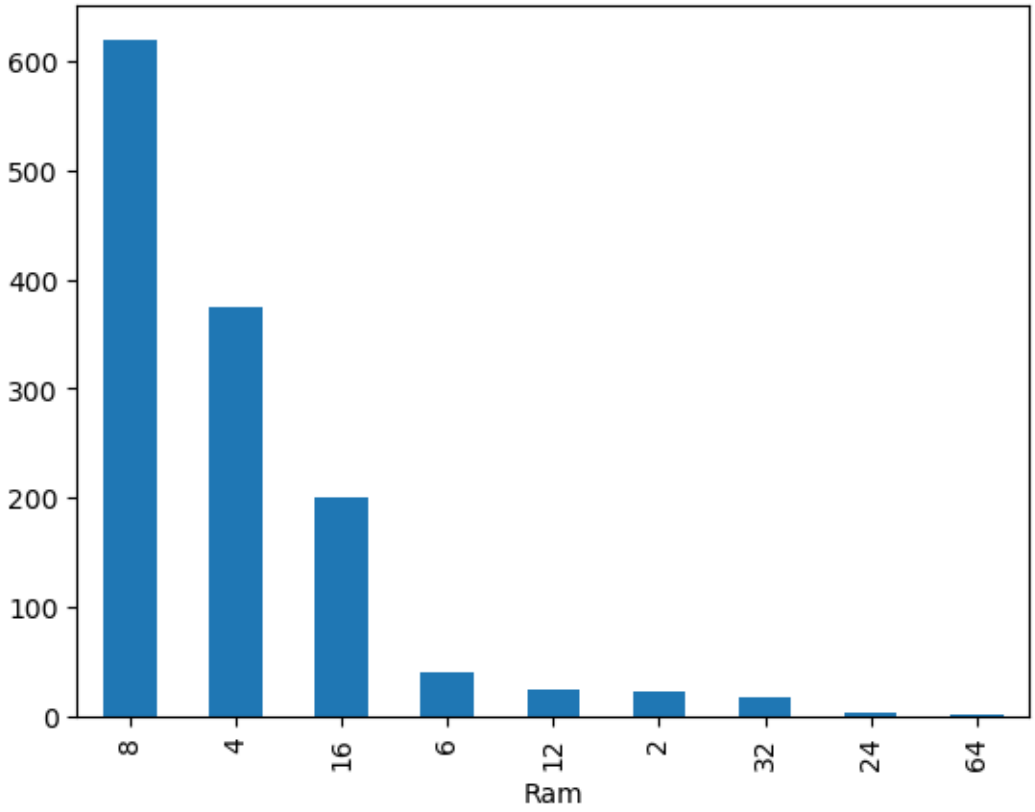
[233]:
```
df.head()
```

[233]:

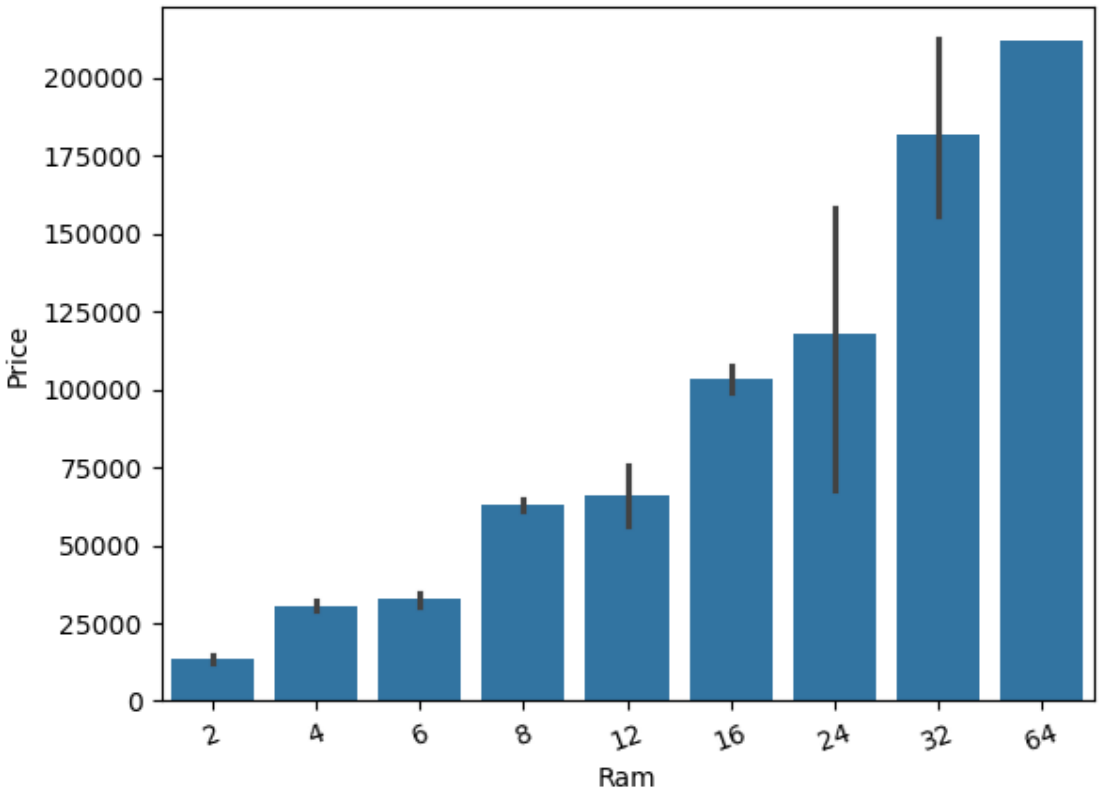
	Company	TypeName	Ram	Memory	Gpu	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu brand
0	Apple	Ultrabook	8	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5
1	Apple	Ultrabook	8	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5
2	HP	Notebook	8	256GB SSD	Intel HD Graphics 620	No OS	1.86	30636.0000	0	0	141.211998	Intel Core i5
3	Apple	Ultrabook	16	512GB SSD	AMD Radeon Pro 455	macOS	1.83	135195.3360	0	1	220.534624	Intel Core i7
4	Apple	Ultrabook	8	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37	96095.8080	0	1	226.983005	Intel Core i5

[234]:
```
df['Ram'].value_counts().plot(kind='bar')
```

[234]: <AxesSubplot: xlabel='Ram'>



[235]:
```
sns.barplot(x=df['Ram'],y=df['Price'])
plt.xticks(rotation=20)
plt.show()
```



[236]:
```
df['Memory'].value_counts()
```

[236]:
```
Memory
256GB SSD                        412
1TB HDD                          223
500GB HDD                        132
512GB SSD                        118
128GB SSD +  1TB HDD              94
128GB SSD                         76
256GB SSD +  1TB HDD              73
32GB Flash Storage                38
2TB HDD                           16
64GB Flash Storage                15
1TB SSD                           14
512GB SSD +  1TB HDD              14
256GB SSD +  2TB HDD              10
1.0TB Hybrid                       9
256GB Flash Storage                8
16GB Flash Storage                 7
32GB SSD                           6
180GB SSD                          5
```

```
128GB Flash Storage                 4
16GB SSD                            3
512GB SSD +  2TB HDD                3
128GB SSD +  2TB HDD                2
256GB SSD +  256GB SSD              2
512GB Flash Storage                 2
1TB SSD +  1TB HDD                  2
256GB SSD +  500GB HDD              2
64GB SSD                            1
512GB SSD +  512GB SSD              1
64GB Flash Storage +  1TB HDD       1
1TB HDD +  1TB HDD                  1
512GB SSD +  256GB SSD              1
32GB HDD                            1
128GB HDD                           1
240GB SSD                           1
8GB SSD                             1
508GB Hybrid                        1
1.0TB HDD                           1
512GB SSD +  1.0TB Hybrid           1
256GB SSD +  1.0TB Hybrid           1
Name: count, dtype: int64
```

[237]:
```
df['Memory'] = df['Memory'].astype(str).replace('\.0', '', regex=True)
df["Memory"] = df["Memory"].str.replace('GB', '')
df["Memory"] = df["Memory"].str.replace('TB', '000')

new = df["Memory"].str.split("+", n=1, expand=True)
df["first"] = new[0].str.strip()
df["second"] = new[1]

df["Layer1HDD"] = df["first"].apply(lambda x: 1 if "HDD" in x else 0)
df["Layer1SSD"] = df["first"].apply(lambda x: 1 if "SSD" in x else 0)
df["Layer1Hybrid"] = df["first"].apply(lambda x: 1 if "Hybrid" in x else 0)
df["Layer1Flash_Storage"] = df["first"].apply(lambda x: 1 if "Flash Storage" in␣
 ↪x else 0)

df['first'] = df['first'].str.extract('(\d+)')
df['first'] = df['first'].fillna(0).astype(int)

df["second"] = df["second"].fillna("0")

df["Layer2HDD"] = df["second"].apply(lambda x: 1 if isinstance(x, str) and␣
 ↪"HDD" in x else 0)
df["Layer2SSD"] = df["second"].apply(lambda x: 1 if isinstance(x, str) and␣
 ↪"SSD" in x else 0)
```

```
df["Layer2Hybrid"] = df["second"].apply(lambda x: 1 if isinstance(x, str) and␣
 ↪"Hybrid" in x else 0)
df["Layer2Flash_Storage"] = df["second"].apply(lambda x: 1 if isinstance(x,␣
 ↪str) and "Flash Storage" in x else 0)

df['second'] = df['second'].str.extract('(\d+)')  # Extract only numeric values
df['second'] = df['second'].fillna(0).astype(int)  # Convert to integer safely

df["HDD"] = (df["first"] * df["Layer1HDD"] + df["second"] * df["Layer2HDD"])
df["SSD"] = (df["first"] * df["Layer1SSD"] + df["second"] * df["Layer2SSD"])
df["Hybrid"] = (df["first"] * df["Layer1Hybrid"] + df["second"] *␣
 ↪df["Layer2Hybrid"])
df["Flash_Storage"] = (df["first"] * df["Layer1Flash_Storage"] + df["second"] *␣
 ↪df["Layer2Flash_Storage"])

df.drop(columns=['first', 'second', 'Layer1HDD', 'Layer1SSD', 'Layer1Hybrid',
                 'Layer1Flash_Storage', 'Layer2HDD', 'Layer2SSD',␣
 ↪'Layer2Hybrid',
                 'Layer2Flash_Storage'], inplace=True)
```

```
<>:1: SyntaxWarning: invalid escape sequence '\.'
<>:17: SyntaxWarning: invalid escape sequence '\d'
<>:28: SyntaxWarning: invalid escape sequence '\d'
<>:1: SyntaxWarning: invalid escape sequence '\.'
<>:17: SyntaxWarning: invalid escape sequence '\d'
<>:28: SyntaxWarning: invalid escape sequence '\d'
/tmp/ipykernel_75163/1007893785.py:1: SyntaxWarning: invalid escape sequence
'\.'
  df['Memory'] = df['Memory'].astype(str).replace('\.0', '', regex=True)
/tmp/ipykernel_75163/1007893785.py:17: SyntaxWarning: invalid escape sequence
'\d'
  df['first'] = df['first'].str.extract('(\d+)')  # Extract only numbers
/tmp/ipykernel_75163/1007893785.py:28: SyntaxWarning: invalid escape sequence
'\d'
  df['second'] = df['second'].str.extract('(\d+)')  # Extract only numeric
values
```

[241]:
```
df.sample()
```

[241]:

	Company	TypeName	Ram	Memory	Gpu	\
153	MSI	Gaming	16	256 SSD + 1000 HDD	Nvidia GeForce GTX 1060	

	OpSys	Weight	Price	Touchscreen	Ips	ppi	\
153	Windows 10	2.8	100699.2	0	0	127.335675	

	Cpu brand	HDD	SSD	Hybrid	Flash_Storage
153	Intel Core i7	1000	256	0	0

[242]:
```
df.drop(columns=['Memory'],inplace=True)
```

[243]:
```
df.head()
```

[243]:
```
  Company   TypeName  Ram                           Gpu  OpSys  Weight  \
0   Apple  Ultrabook    8  Intel Iris Plus Graphics 640  macOS    1.37
1   Apple  Ultrabook    8        Intel HD Graphics 6000  macOS    1.34
2      HP   Notebook    8         Intel HD Graphics 620  No OS    1.86
3   Apple  Ultrabook   16            AMD Radeon Pro 455  macOS    1.83
4   Apple  Ultrabook    8  Intel Iris Plus Graphics 650  macOS    1.37

         Price  Touchscreen  Ips         ppi      Cpu brand  HDD  SSD  Hybrid  \
0   71378.6832            0    1  226.983005  Intel Core i5    0  128       0
1   47895.5232            0    0  127.677940  Intel Core i5    0    0       0
2   30636.0000            0    0  141.211998  Intel Core i5    0  256       0
3  135195.3360            0    1  220.534624  Intel Core i7    0  512       0
4   96095.8080            0    1  226.983005  Intel Core i5    0  256       0

   Flash_Storage
0              0
1            128
2              0
3              0
4              0
```

[245]:
```
df.select_dtypes(include='number').corr()['Price']
```

[245]:
```
Ram              0.743007
Weight           0.210370
Price            1.000000
Touchscreen      0.191226
Ips              0.252208
ppi              0.473487
HDD             -0.096441
SSD              0.670799
Hybrid           0.007989
Flash_Storage   -0.040511
Name: Price, dtype: float64
```

[246]:
```
df.drop(columns=['Hybrid','Flash_Storage'],inplace=True)
```

[247]:
```
df.head()
```

[247]:
```
  Company   TypeName  Ram                           Gpu  OpSys  Weight  \
0   Apple  Ultrabook    8  Intel Iris Plus Graphics 640  macOS    1.37
1   Apple  Ultrabook    8        Intel HD Graphics 6000  macOS    1.34
2      HP   Notebook    8         Intel HD Graphics 620  No OS    1.86
3   Apple  Ultrabook   16            AMD Radeon Pro 455  macOS    1.83
```

```
4   Apple  Ultrabook    8  Intel Iris Plus Graphics 650  macOS    1.37

         Price  Touchscreen  Ips         ppi      Cpu brand  HDD  SSD
0   71378.6832            0    1  226.983005  Intel Core i5    0  128
1   47895.5232            0    0  127.677940  Intel Core i5    0    0
2   30636.0000            0    0  141.211998  Intel Core i5    0  256
3  135195.3360            0    1  220.534624  Intel Core i7    0  512
4   96095.8080            0    1  226.983005  Intel Core i5    0  256
```

[248]:
```
df['Gpu'].value_counts()
```

[248]:
```
Gpu
Intel HD Graphics 620      281
Intel HD Graphics 520      185
Intel UHD Graphics 620      68
Nvidia GeForce GTX 1050     66
Nvidia GeForce GTX 1060     48
                          ...
AMD Radeon R7 Graphics       1
AMD Radeon R7 M360           1
Nvidia Quadro M3000M         1
Nvidia GeForce 960M          1
ARM Mali T860 MP4            1
Name: count, Length: 110, dtype: int64
```

[249]:
```
df['Gpu brand'] = df['Gpu'].apply(lambda x:x.split()[0])
```

[250]:
```
df.head()
```

[250]:
```
  Company   TypeName  Ram                           Gpu  OpSys  Weight  \
0   Apple  Ultrabook    8  Intel Iris Plus Graphics 640  macOS    1.37
1   Apple  Ultrabook    8        Intel HD Graphics 6000  macOS    1.34
2      HP   Notebook    8         Intel HD Graphics 620  No OS    1.86
3   Apple  Ultrabook   16            AMD Radeon Pro 455  macOS    1.83
4   Apple  Ultrabook    8  Intel Iris Plus Graphics 650  macOS    1.37

         Price  Touchscreen  Ips         ppi      Cpu brand  HDD  SSD  \
0   71378.6832            0    1  226.983005  Intel Core i5    0  128
1   47895.5232            0    0  127.677940  Intel Core i5    0    0
2   30636.0000            0    0  141.211998  Intel Core i5    0  256
3  135195.3360            0    1  220.534624  Intel Core i7    0  512
4   96095.8080            0    1  226.983005  Intel Core i5    0  256

  Gpu brand
0     Intel
1     Intel
2     Intel
```

```
3       AMD
4     Intel
```

[251]:
```
df['Gpu brand'].value_counts()
```

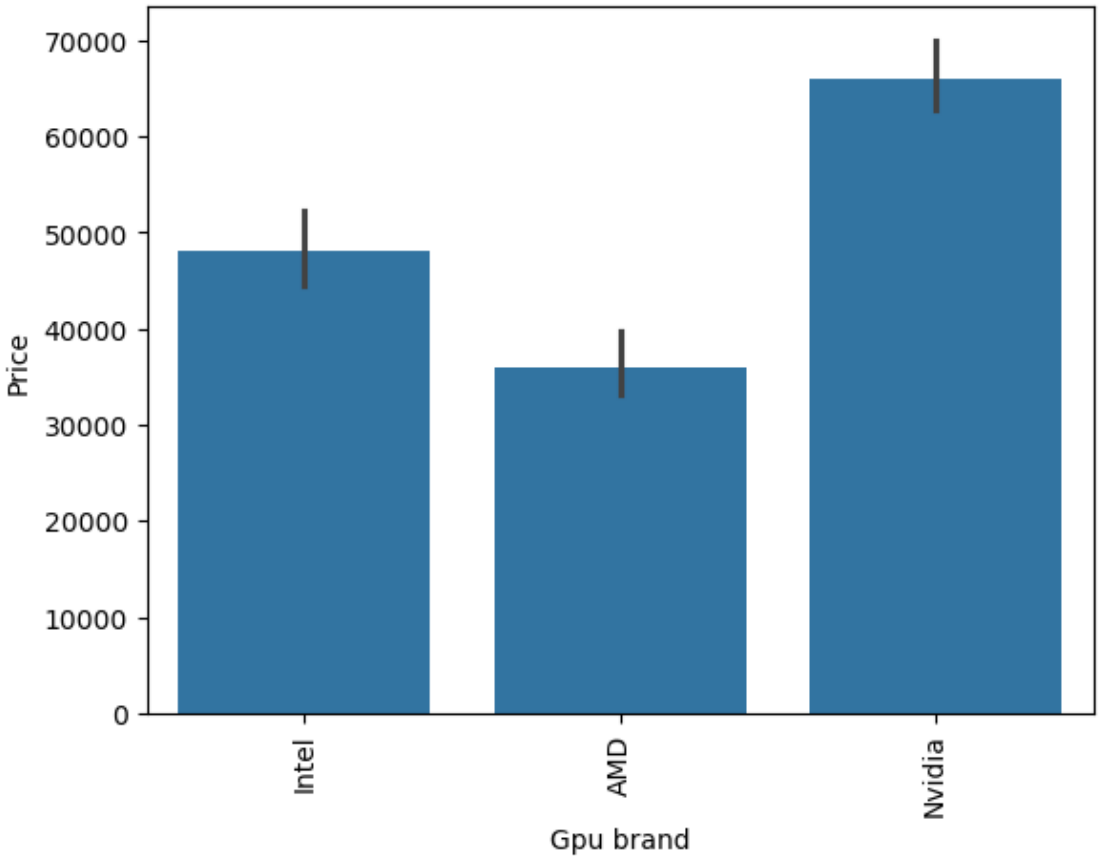
[251]:
```
Gpu brand
Intel     722
Nvidia    400
AMD       180
ARM         1
Name: count, dtype: int64
```

[252]:
```
df = df[df['Gpu brand'] != 'ARM']
```

[253]:
```
df['Gpu brand'].value_counts()
```

[253]:
```
Gpu brand
Intel     722
Nvidia    400
AMD       180
Name: count, dtype: int64
```

[254]:
```
sns.barplot(x=df['Gpu brand'],y=df['Price'],estimator=np.median)
plt.xticks(rotation='vertical')
plt.show()
```



[255]:
```
df.drop(columns=['Gpu'],inplace=True)
```

```
/tmp/ipykernel_75163/1111925144.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df.drop(columns=['Gpu'],inplace=True)
```

[256]:
```
df.head()
```

[256]:

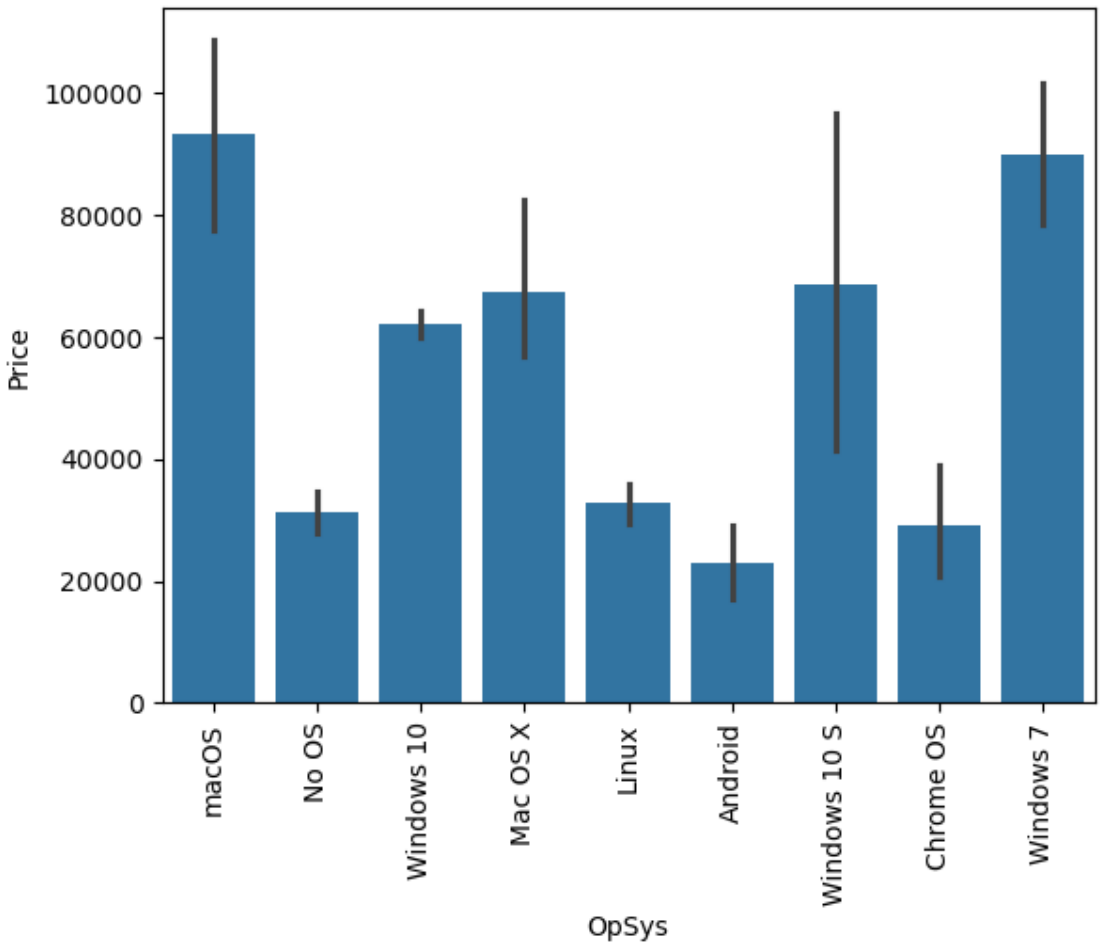
	Company	TypeName	Ram	OpSys	Weight	Price	Touchscreen	Ips	ppi	Cpu brand	HDD	SSD	Gpu brand
0	Apple	Ultrabook	8	macOS	1.37	71378.6832	0	1	226.983005	Intel Core i5	0	128	Intel
1	Apple	Ultrabook	8	macOS	1.34	47895.5232	0	0	127.677940	Intel Core i5	0	0	Intel
2	HP	Notebook	8	No OS	1.86	30636.0000	0	0	141.211998	Intel Core i5	0	256	Intel
3	Apple	Ultrabook	16	macOS	1.83	135195.3360	0	1	220.534624	Intel Core i7	0	512	AMD
4	Apple	Ultrabook	8	macOS	1.37	96095.8080	0	1	226.983005	Intel Core i5	0	256	Intel

```
1  127.677940  Intel Core i5    0    0     Intel
2  141.211998  Intel Core i5    0  256     Intel
3  220.534624  Intel Core i7    0  512       AMD
4  226.983005  Intel Core i5    0  256     Intel
```

[257]:
```
df['OpSys'].value_counts()
```

[257]:
```
OpSys
Windows 10      1072
No OS             66
Linux             62
Windows 7         45
Chrome OS         26
macOS             13
Mac OS X           8
Windows 10 S       8
Android            2
Name: count, dtype: int64
```

[258]:
```
sns.barplot(x=df['OpSys'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```



[259]:
```
def cat_os(inp):
    if inp == 'Windows 10' or inp == 'Windows 7' or inp == 'Windows 10 S':
        return 'Windows'
    elif inp == 'macOS' or inp == 'Mac OS X':
        return 'Mac'
    else:
        return 'Others/No OS/Linux'
```

[260]:
```
df['os'] = df['OpSys'].apply(cat_os)
```

```
/tmp/ipykernel_75163/3648919379.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df['os'] = df['OpSys'].apply(cat_os)
```

[261]:
```
df.head()
```

[261]:

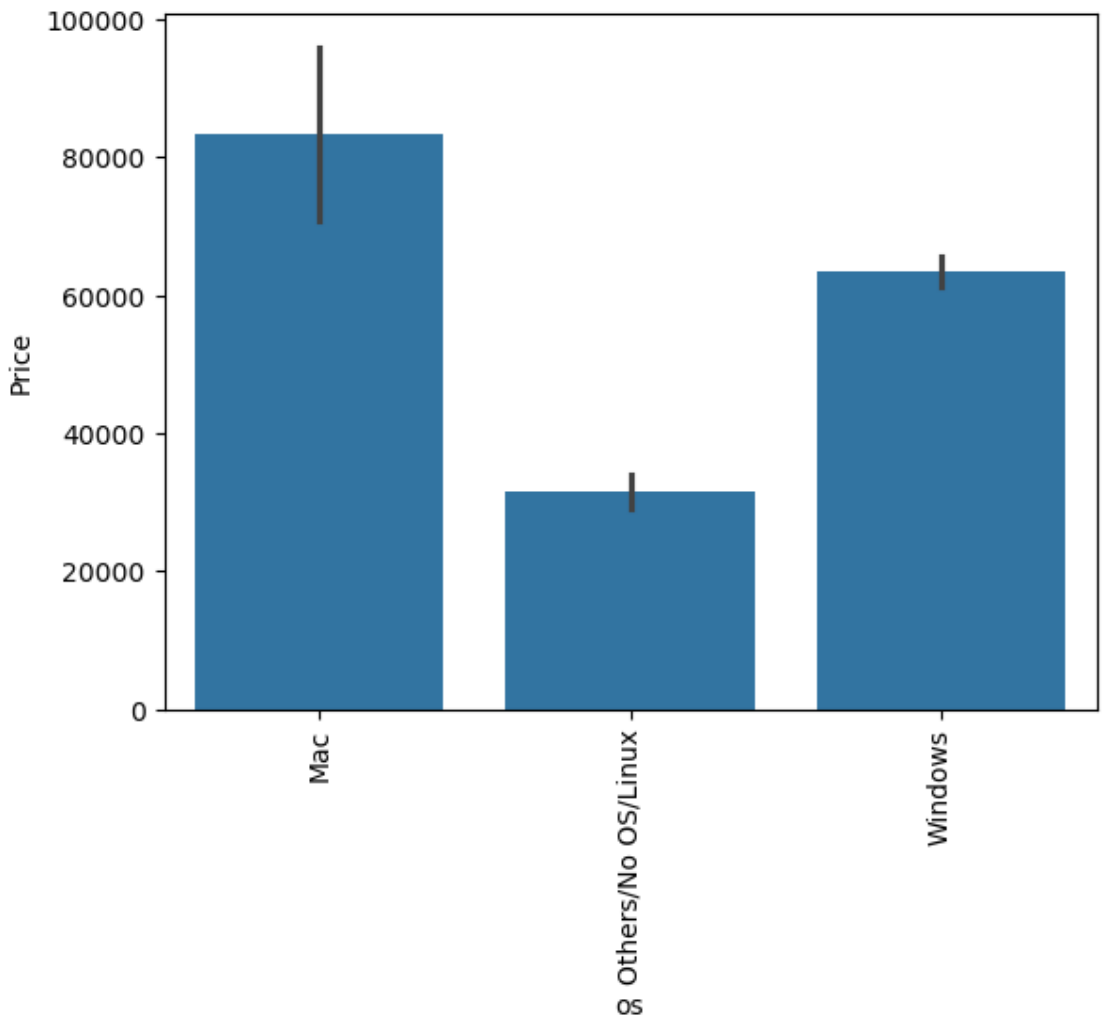
	Company	TypeName	Ram	OpSys	Weight	Price	Touchscreen	Ips	\
0	Apple	Ultrabook	8	macOS	1.37	71378.6832	0	1	
1	Apple	Ultrabook	8	macOS	1.34	47895.5232	0	0	
2	HP	Notebook	8	No OS	1.86	30636.0000	0	0	
3	Apple	Ultrabook	16	macOS	1.83	135195.3360	0	1	
4	Apple	Ultrabook	8	macOS	1.37	96095.8080	0	1	

	ppi	Cpu brand	HDD	SSD	Gpu brand	os
0	226.983005	Intel Core i5	0	128	Intel	Mac
1	127.677940	Intel Core i5	0	0	Intel	Mac
2	141.211998	Intel Core i5	0	256	Intel	Others/No OS/Linux
3	220.534624	Intel Core i7	0	512	AMD	Mac
4	226.983005	Intel Core i5	0	256	Intel	Mac

[262]:
```
df.drop(columns=['OpSys'],inplace=True)
```

```
/tmp/ipykernel_75163/3105339334.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df.drop(columns=['OpSys'],inplace=True)
```

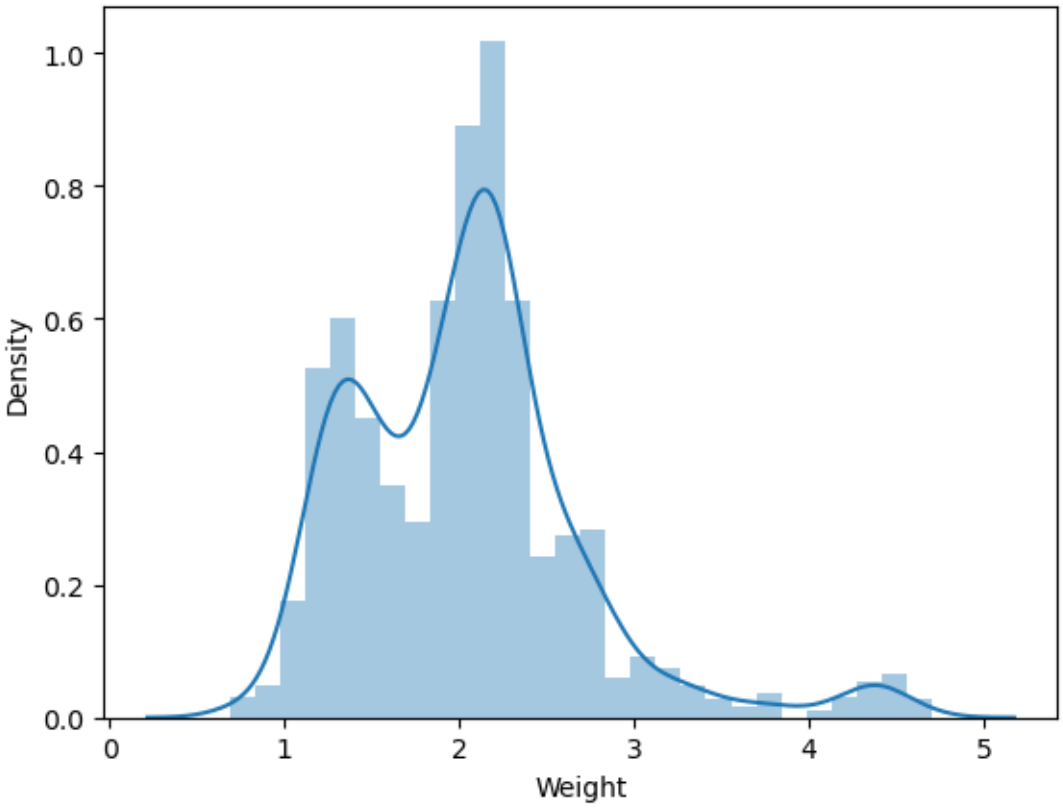
[263]:
```
sns.barplot(x=df['os'],y=df['Price'])
plt.xticks(rotation='vertical')
plt.show()
```



[265]:
```
sns.distplot(df['Weight'])
```

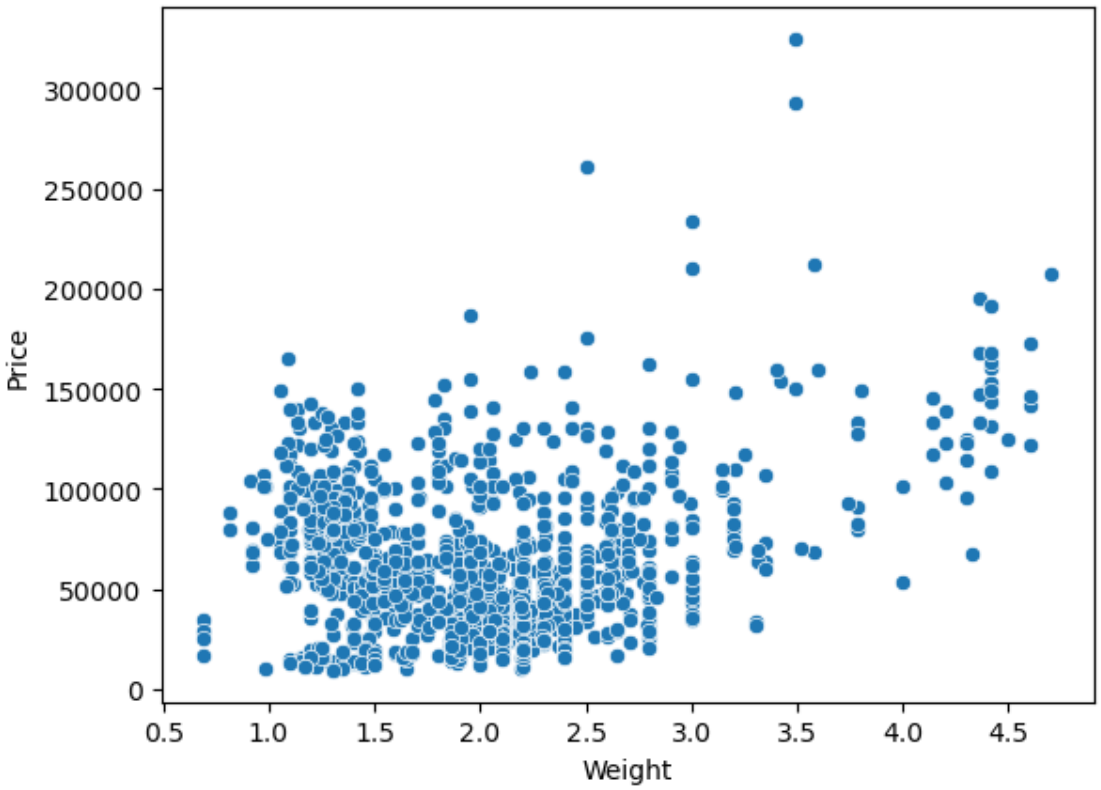
```
/tmp/ipykernel_75163/1125578356.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['Weight'])
```

[265]: <AxesSubplot: xlabel='Weight', ylabel='Density'>



[266]:
```
sns.scatterplot(x=df['Weight'],y=df['Price'])
```

[266]: <AxesSubplot: xlabel='Weight', ylabel='Price'>

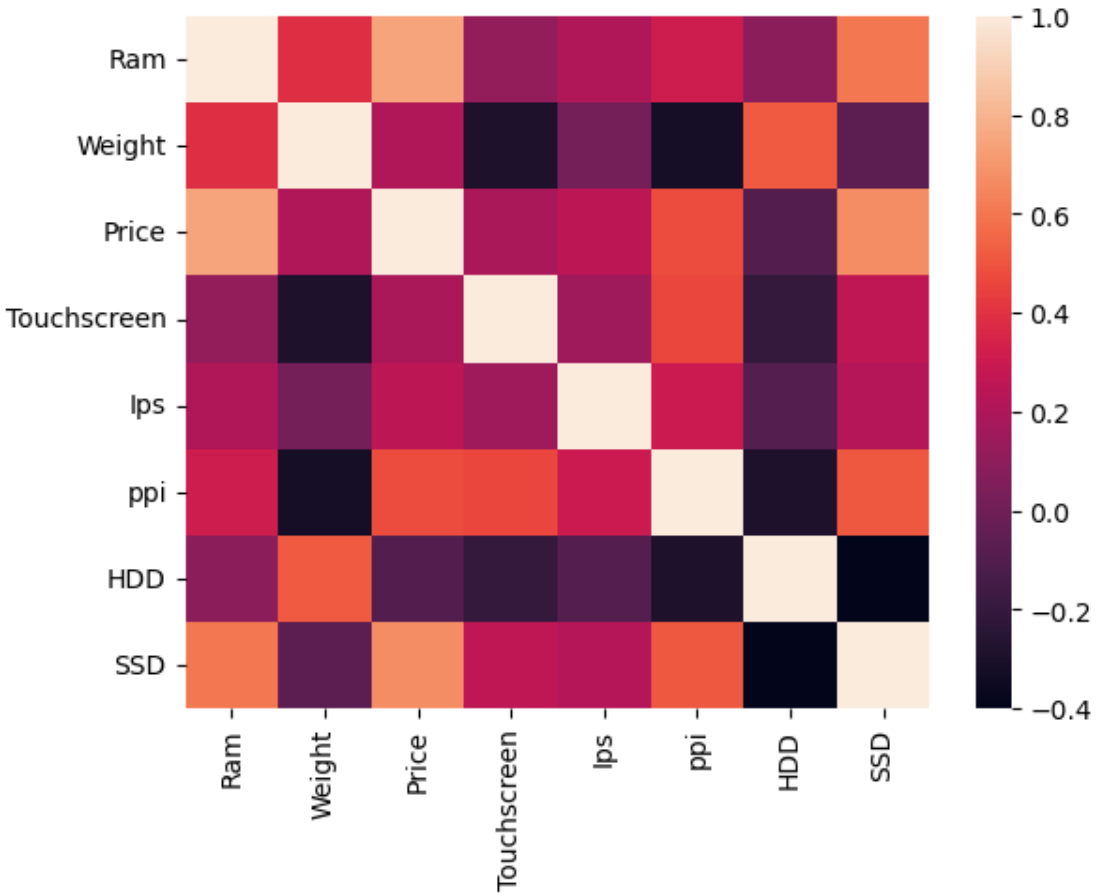


[267]:
```
df.select_dtypes(include='number').corr()['Price']
```

[267]:
```
Ram            0.742905
Weight         0.209867
Price          1.000000
Touchscreen    0.192917
Ips            0.253320
ppi            0.475368
HDD           -0.096891
SSD            0.670660
Name: Price, dtype: float64
```

[269]:
```
sns.heatmap(df.select_dtypes(include='number').corr())
```

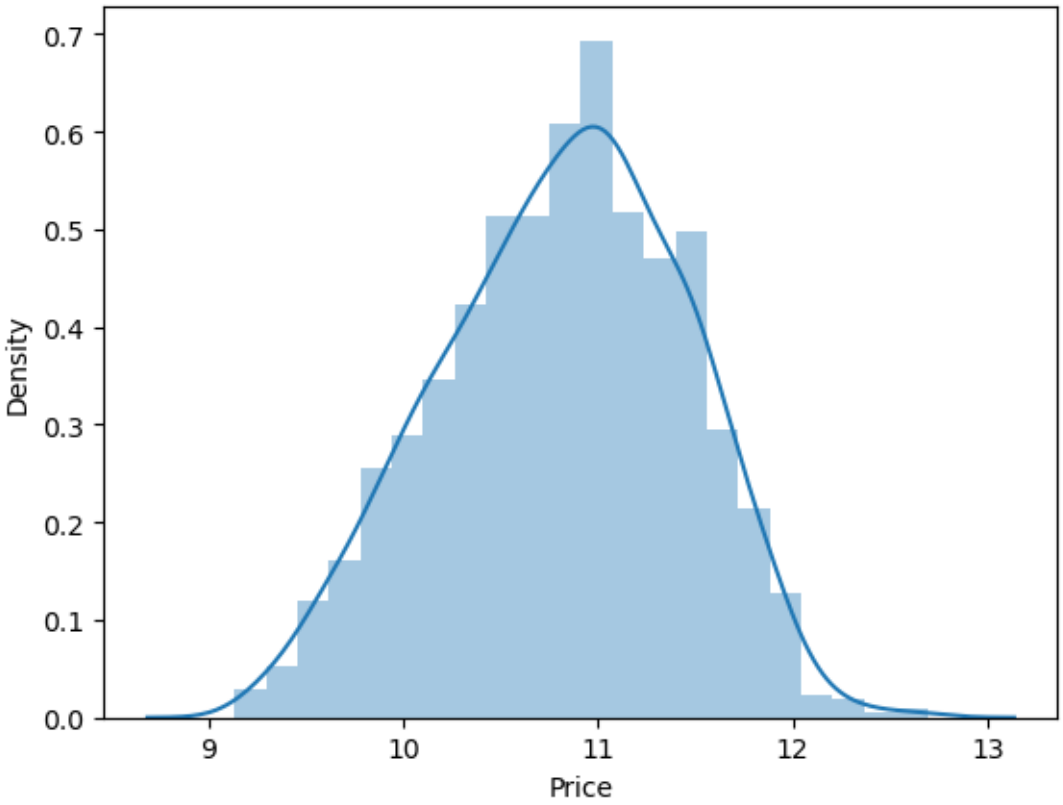
[269]:
```
<AxesSubplot: >
```



[270]:
```
sns.distplot(np.log(df['Price']))
```

```
/tmp/ipykernel_75163/3556049916.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(np.log(df['Price']))
```

[270]: `<AxesSubplot: xlabel='Price', ylabel='Density'>`



[276]:
```
X = df.drop(columns=['Price'])
y = np.log(df['Price'])
```

[305]:
```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.
 ↪15,random_state=2)
```

[306]:
```
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder
from sklearn.metrics import r2_score,mean_absolute_error
```

[307]:
```
from sklearn.ensemble import RandomForestRegressor
```

[413]:
```
col_transformer = ColumnTransformer(
    transformers=[
        ('onehot', OneHotEncoder(sparse_output=False, drop='first'), [0, 1, 7,␣
 ↪10, 11])
    ],
    remainder='passthrough'
)
```

```
rf_regressor = RandomForestRegressor(
    n_estimators=74,
    random_state=5,
    max_samples=0.8,
    max_features=1,
    max_depth=19,
)

pipe = Pipeline([
    ('preprocessor', col_transformer),
    ('regressor', rf_regressor)
])


pipe.fit(X_train, y_train)

y_pred = pipe.predict(X_test)

print('R2 Score:',r2_score(y_test, y_pred))
print('MAE:', mean_absolute_error(y_test, y_pred))
```

```
R2 Score: 0.9094921614316613
MAE: 0.1419151635099536
```

[414]:
```
import pickle
```

[415]:
```
pickle.dump(df,open('df.pkl','wb'))
pickle.dump(pipe,open('pipe.pkl','wb'))
```

[ ]:
```

```